Heart Disease Prediction Analysis

By: Joshua Kim

Table of Contents

Preparation 3
- **Link to Dataset** 3
- **Variables and Types** 3

Supervised Learning with Decision Trees 7
- **Decision Tree Preparation** 7
- **Decision Tree and Confusion Matrix** 9
- **Conclusion** 11

Unsupervised Learning with K-Means Clustering 12
- **K-Means Preparation** 12
- **K-Means Calculation** 13
- **Conclusion** 14

Exploratory Data Analysis with SQL 15
- **Patient Demographic Attributes** 15
- **Patient Anthropometric Measurements** 17
- **Patient Symptoms and Heart Disease** 20
- **Patient Metabolic Markers** 21
- **Midway Conclusions** 22
- **Multiple Variables Analysis** 23

Final Conclusions 25

Preparation

Link to Data Set: https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction?resource=download

Variables and Types

1. Age
    a. Type: Qualitative
    b. Description: The age of the patient in years.
2. Sex
    a. Type: Qualitative
    b. Description: The sex of the patient. In this dataset 'M' represents Male and 'F' represents Female.
3. ChestPainType
    a. Type: Qualitative
    b. Description:
        i. This dataset consists of four pain types: TA ("Typical Angina"), ATA ("Atypical Angina"), NAP ("Non-Anginal Pain"), and ASY ("Asymptomatic")
4. RestingBP
    a. Type: Quantitative
    b. Description: The resting [systolic] blood pressure of the patient in mmHg
5. Cholesterol
    a. Type: Quantitative
    b. Description: The serum cholesterol levels of a patient in mm/dL
6. FastingBS
    a. Type: Qualitative
    b. Description: Depicts the fasting blood sugar levels of a patient, with 0 being under 120 mm/dL and 1 being otherwise.
7. RestingECG
    a. Type: Qualitative
    b. Description:
        i. This dataset consists of 3 Electrocardiogram reading categories: Normal, ST (ST-T wave abnormality), and LVH(possible or present left ventricular hypertrophy, or enlargement of left ventricle of the heart.)
8. MaxHR
    a. Type: Quantitative
    b. Description: The maximum heart rate of the patient.
9. ExerciseAnginga
    a. Type: Qualitative
    b. Description: Depicts the presence of chest pain during exercise.
10. Oldpeak
    a. Type: Quantitative
    b. Description: Numerical value measurement of ST segment depression in a patient.
11. ST_Slope
    a. Type: Qualitative
    b. Description: The slope of the ST segment on an ECG reading, often during exercise tests. Depicts the relationship between the ST segment and the heart rate. Three categories: Up, Down, and Flat.
12. HeartDisease
    a. Type: Dataset Label
    b. Names: 0, 1
    c. Description: This is the target variable of the dataset, as heart disease is the leading cause of heart failure. In this dataset, 0 depicts no heart disease, while 1 depicts the presence of heart disease.

Dataset Cleaning

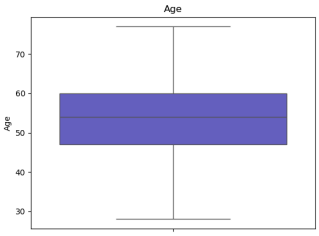
In this section, I will be cleaning the Heart Dataset. In its current state, the dataset contains invalid fields in the RestingBP (resting blood pressure) and Cholesterol fields. In these fields, there are 0 values which should not be possible and thus will be removed. In order for proper data analysis to be conducted, these values must be cleaned. All other fields in this dataset consisted of normal, reasonable, or possible values, so they are considered clean. In the following pages, I will be detailing the process of cleaning this dataset so that it is suited for analysis.

The following image depicts the invalid state of the dataset:

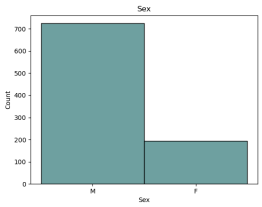
Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
55	M	NAP	0	0	0	Normal	155	N	1.5	Flat	1
53	M	ASY	80	0	0	Normal	141	Y	2	Down	0
38	M	ASY	92	117	0	Normal	134	Y	2.5	Flat	1
39	F	NAP	94	199	0	Normal	179	N	0	Up	0
51	M	NAP	94	227	0	Normal	154	Y	0	Up	0
32	M	TA	95	0	1	Normal	127	N	0.7	Up	1
51	M	ASY	95	0	1	Normal	126	N	2.2	Flat	1
57	M	ASY	95	0	1	Normal	182	N	0.7	Down	1
52	M	ASY	95	0	1	Normal	82	Y	0.8	Flat	1
40	M	ASY	95	0	1	ST	144	N	0	Up	1
64	F	ASY	95	0	1	Normal	145	N	1.1	Down	1
63	M	ASY	96	305	0	ST	121	Y	1	Up	1
34	M	ATA	98	220	0	Normal	150	N	0	Up	0
63	M	ASY	100	0	1	Normal	109	N	-0.9	Flat	1
46	M	ASY	100	0	1	ST	133	N	-2.6	Flat	1
38	M	NAP	100	0	0	Normal	179	N	-1.1	Up	0
43	M	ASY	100	0	1	Normal	122	N	1.5	Down	1
48	M	ATA	100	159	0	Normal	100	N	0	Up	0
58	M	ASY	100	213	0	ST	110	N	0	Up	0
31	F	ATA	100	219	0	ST	150	N	0	Up	0
51	M	NAP	100	222	0	Normal	143	Y	1.2	Flat	0
43	F	TA	100	223	0	Normal	142	N	0	Up	0
58	M	ASY	100	234	0	Normal	156	N	0.1	Up	1
33	F	ASY	100	246	0	Normal	150	Y	1	Flat	1
60	M	ASY	100	248	0	Normal	125	N	1	Flat	1
58	F	ASY	100	248	0	LVH	122	N	1	Flat	0
49	M	ATA	100	253	0	Normal	174	N	0	Up	0
67	M	ASY	100	299	0	LVH	125	Y	0.9	Flat	1
46	M	ATA	101	197	1	Normal	156	N	0	Up	0
48	M	NAP	102	0	1	ST	110	Y	1	Down	1
42	F	ASY	102	265	0	LVH	122	N	0.6	Flat	0
60	F	NAP	102	318	0	Normal	160	N	0	Up	0
41	M	ASY	104	0	0	ST	111	N	0	Up	0

Variables to be cleaned:

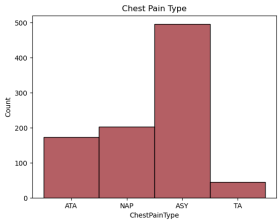
1. Age:
   a. The age field is a quantitative field that could reveal important correlations in terms of heart disease and failure risk, so it is retained in the dataset.
   b. According to the visualization on the right, there are no outliers present in the set. There are also no nonsensical values.
   c. Thus, no extra data cleaning was required on this field.



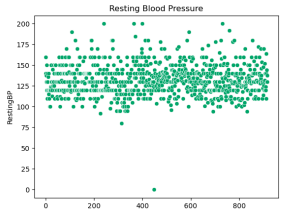
2. Sex:
   a. The sex field is a qualitative field that also could reveal important correlations such as whether a male or female is more likely to be at risk of heart disease.
   b. According to the visualization, there are no nonsensical values in this set, and so no extra data cleaning was required.



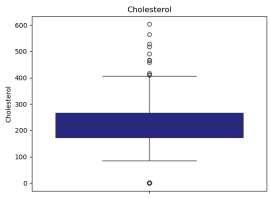
3. ChestPainType:
   a. The chest pain type field is a qualitative field that describes the type of chest pain a patient has, and could reveal which type of pain could more likely be indicative of heart disease.
   b. This field was also already clean, as there were no nonsensical values or value counts considered outliers.

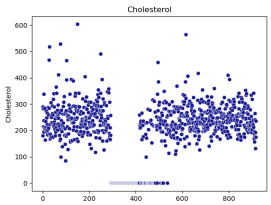


4. RestingBP:
   a. This field is one that displays all the records of each patient's systolic blood pressure reading, and could reveal trends between blood pressure levels and the likelihood of heart disease.
   b. As we can see in the scatter plot, there was one value recorded at 0, which is medically impossible.
   c. To clean this field, the 0 value was removed along with the row that it was contained in.

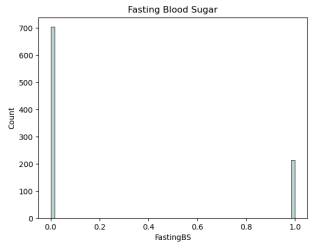


5. Cholesterol:
   a. The cholesterol field records the serum cholesterol levels in each patient, which could help reveal whether there are trends with rising cholesterol levels and risk for heart disease.
   b. From the boxplot, we can see there are several outliers. However, viewing the data in this field through a scatterplot, we can see that the high outliers are not so far removed from the dataset, hence these values were kept.
   c. However, the zero values in the cholesterol field were removed, as this condition is extremely rare.

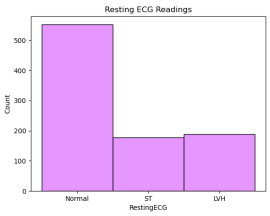




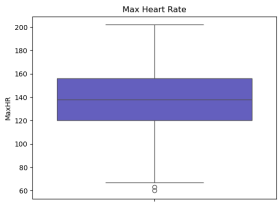
6. FastingBS:
   a. The fasting blood sugar field is another important field in terms of identifying factors of heart disease risk and so the field as a whole was not removed.
   b. From the histogram, we can see that there are no outlier counts and no nonsensical values, and so this field is considered already clean and no further preparation was required.

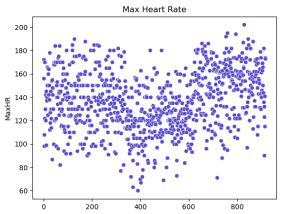


7. RestingECG:
    a. This field is another important field that can show whether a certain ECG reading is more prone to be paired with heart disease presence in a patient.
    b. From this histogram, we can see that there are also no outlier counts and no nonsensical values.
    c. Hence, no further preparation was required for this field as well.

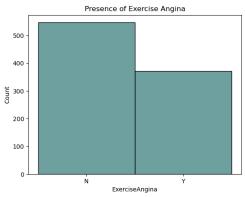


8. MaxHR:
    a. The maximum heart rate field could reveal if there were any important correlations between maximum heart rate recorded in a patient.
    b. From the boxplot, we can see that there are a couple outliers towards the lower end of the value range. However, when running a visualization through a scatterplot, we can see that these values are not far removed from the rest of the dataset, and so were kept in the dataset.

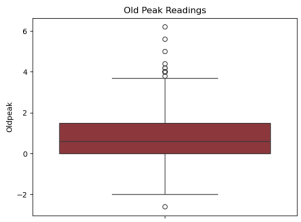




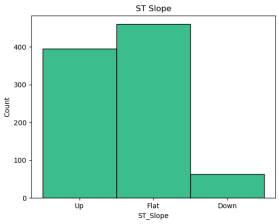
9. ExerciseAngina:
    a. The presence of exercise angina could reveal whether someone who has heart disease usually presents with this symptom, and thus revealing any correlations between the two.
    b. From this histogram, we can see there are no outlier value counts and no nonsensical values. Hence, this field is considered clean and no preparation or manipulation was required.



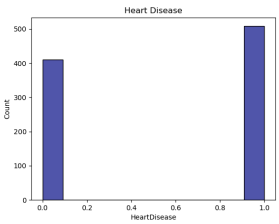
10. Oldpeak:
    a. The Oldpeak field is one that should have direct correlation to a patient's ST slope, and so was kept in the dataset to aid deeper and precise analysis.
    b. According to the boxplot, there are some outliers, but as with a few of the previous fields, I ran another visualization through a scatter plot. In the scatter plot, again we can see that these values are not too removed from the rest of the dataset, so I chose to retain them in the set.



11. ST_Slope:
    a. The ST slope field is one that could provide insight into an individual's risk for heart disease as well as supplement fields such as the Oldpeak field and the RestingECG field, and so it was not removed from the dataset.
    b. According to the histogram, there were no outlier value counts and no nonsensical values present, so no further cleaning was necessary on this field.



12. HeartDisease
    a. This field is the label of the dataset, and the target variable for analysis, so it was retained.
    b. This field also contained no nonsensical values or outlier counts, and so no cleaning was necessary.



13. PatientID
    a. After cleaning, this field was added to the dataset for the SQL portion of the analysis, to aid in counting measures.

Supervised Learning with Decision Trees

Decision Tree Preparation:

1. To start off this analysis, I chose to explore the data by first using Supervised Learning, as the purpose of this project is to aid in the prediction and/or diagnosis of Heart Disease and ultimately Heart Failure.

2. To prepare the dataset for this section, I used the cleaned version of the Heart Dataset, as it contained the target variable for the analysis - a factor required for Decision Tree modeling.

3. However, for Decision Tree modeling, all fields must be quantitative or, at the very least, numerical. So I cleaned the dataset further by replacing the values in the 'Sex' field where 'Males' now equaled 0 and 'Females' now equaled 1. I repeated this process for the 'ExerciseAngina' field, where a response of 'No' equaled to 0 and a response of 'Yes' equaled to 1.

4. All other non-numerical attributes that could not be represented as binary values were dropped. These columns were 'RestingECG', 'ST_Slope', and 'ChestPainType'.

5. After this phase of cleaning, I split the dataset into two different sets, one for training the Decision Tree model and one for testing it. This process was done using sklearn's train_test_split function with a test size of 0.2 (80% of the original dataset would be used to train the model and 20% would be used to test it) and a random state of 42 (to maintain a controlled randomness when splitting the set.)
   a. Training set:

```
The training Data is:
      Age  Sex  RestingBP  Cholesterol  FastingBS  MaxHR  ExerciseAngina  Oldpeak  HeartDisease
70    57    0        140          265          0    145               1      1.0             1
164   52    1        140          225          0    140               0      0.0             0
710   56    1        140          294          0    153               0      1.3             0
265   54    0        160          305          0    175               0      0.0             0
250   44    0        135          491          0    135               0      0.0             1
..   ...  ...        ...          ...        ...    ...             ...      ...           ...
71    44    0        130          215          0    135               0      0.0             0
106   48    1        120          254          0    110               0      0.0             0
270   45    0        120          225          0    140               0      0.0             0
435   53    0        144          300          1    128               1      1.5             1
102   40    1        150          392          0    130               0      2.0             1

[596 rows x 9 columns]
```

   b. Testing set:

```
The testing data is:
      Age  Sex  RestingBP  Cholesterol  FastingBS  MaxHR  ExerciseAngina  Oldpeak  HeartDisease
208   28    0        130          132          0    185               0      0.0             0
259   55    1        122          320          0    155               0      0.0             0
97    39    0        160          147          1    160               0      0.0             0
148   50    0        120          168          0    160               0      0.0             0
395   71    0        130          221          0    115               1      0.0             1
..   ...  ...        ...          ...        ...    ...             ...      ...           ...
554   41    0        110          235          0    153               0      0.0             0
69    44    0        150          412          0    170               0      0.0             0
522   56    0        120          236          0    178               0      0.8             0
131   46    0        110          202          0    150               1      0.0             1
44    43    0        120          175          0    120               1      1.0             1

[150 rows x 9 columns]
```

6. I then obtained the counts for each label ('HeartDisease') option for the dataset and found that it was unbalanced. In order to produce as accurate results as possible when training and testing the model, I took steps to balance the training set to produce an equal number of possible label options.

```
Label counts:
 HeartDisease
0    319
1    277
Name: count, dtype: int64
```

```
Balanced data counts:
 HeartDisease
0    277
1    277
Name: count, dtype: int64
```

7. Finally, I removed the labels from the training (balanced) and testing datasets, but retained the labels separately.
   a. Training set:

```
Training set:
      Age  Sex  RestingBP  Cholesterol  FastingBS  MaxHR  ExerciseAngina  Oldpeak
453   59    0        135          234          0    161               0      0.5
178   37    0        130          194          0    150               0      0.0
415   37    0        118          240          0    165               0      1.0
322   57    0        140          214          0    144               1      2.0
468   48    1        130          275          0    139               0      0.2
..   ...  ...        ...          ...        ...    ...             ...      ...
726   35    0        122          192          0    174               0      0.0
48    36    0        112          340          0    184               0      1.0
710   56    1        140          294          0    153               0      1.3
621   67    0        125          254          1    163               0      0.2
292   53    0        130          182          0    148               0      0.0

[554 rows x 8 columns]
```

```
Training Labels:
 453    0
178    0
415    0
322    1
468    0
      ..
726    0
48     0
710    0
621    1
292    0
Name: HeartDisease,
```

   b. Testing set:

```
Testing set:
      Age  Sex  RestingBP  Cholesterol  FastingBS  MaxHR  ExerciseAngina  Oldpeak
208   28    0        130          132          0    185               0      0.0
259   55    1        122          320          0    155               0      0.0
97    39    0        160          147          1    160               0      0.0
148   50    0        120          168          0    160               0      0.0
395   71    0        130          221          0    115               1      0.0
..   ...  ...        ...          ...        ...    ...             ...      ...
554   41    0        110          235          0    153               0      0.0
69    44    0        150          412          0    170               0      0.0
522   56    0        120          236          0    178               0      0.8
131   46    0        110          202          0    150               1      0.0
44    43    0        120          175          0    120               1      1.0

[150 rows x 8 columns]
```

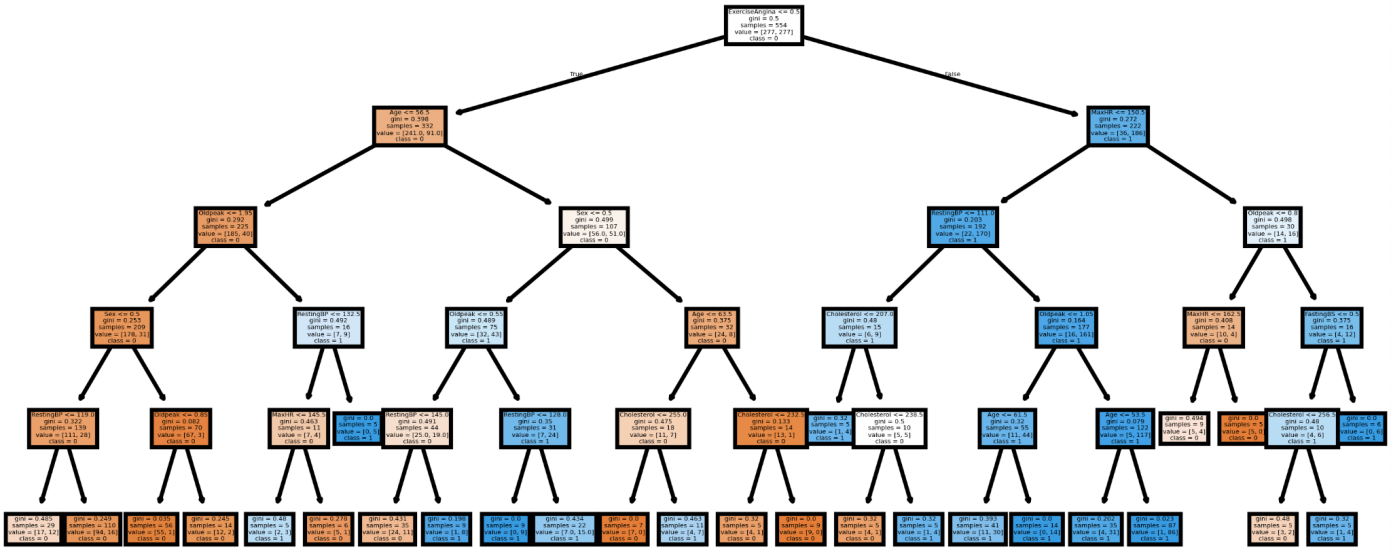
```
Testing labels:
 208    0
259    0
97     0
148    0
395    1
      ..
554    0
69     0
522    0
131    1
44     1
Name: HeartDisease,
```

Decision Tree and Confusion Matrix:

1. Now that the original cleaned dataset has been split between a training and testing portion, with the training dataset being balanced, and with the labels separated and retained, we are ready to train and test the Decision Tree model and test the results using a confusion matrix.

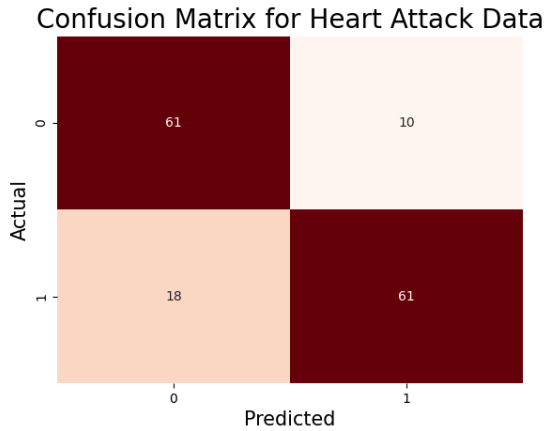
2. For the purpose of this analysis, the Decision Tree was created with the following attributes:
    a. Max depth of 4 levels
    b. Minimum sample split of 5
        i. This denotes the minimum number of samples required in a node before the algorithm is allowed to split the node.
        ii. In simpler terms: determines whether the node can be split.
    c. Minimum sample leaf of 3
        i. This denotes the minimum number of samples that must remain in each leaf node after a split.
        ii. In simpler terms: determines whether the resulting leaf will be large enough with sample size.

3. With the attributes listed above, this was the resulting Decision Tree after fitting it with my training dataset and training labels.



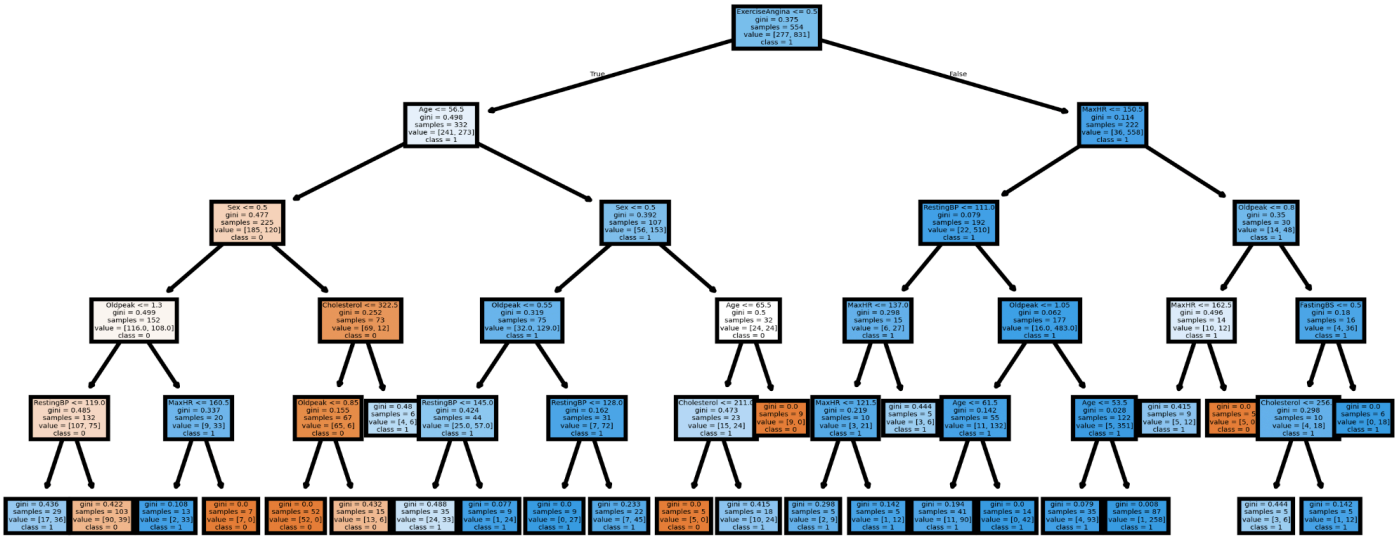
    a. This Decision Tree model depicts a certain order of importances: ExerciseAngina -> Age -> Oldpeak -> RestingBP (blood pressure) -> MaxHR (heart rate) -> Sex -> Cholesterol -> FastingBS (blood sugar)
    b. This was the pathway the model took in determining whether a patient had heart disease or not.

4. I then tested this model for its accuracy by creating a prediction for the Testing Dataset then tested the accuracy of this model using a confusion matrix.



   a. In this model, we can see that there is a good number of true positive and true negative matches, highlighting that this model is fairly accurate. However, there are 18 cases of false negatives, which, in a clinical setting, is not ideal.

5. I decided to edit the Decision Tree classifier by changing the class weight attribute of the classifier from a balanced weight to a custom weight. The custom weight I used was: {0: 1, 1: 3}. After doing so, I reran the code again to produce another tree:

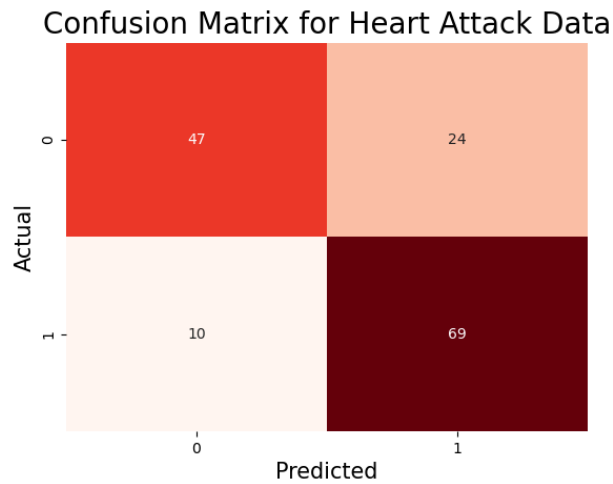


6. I tested this model again by creating a new prediction, which was as follows:

```
Prediction for testing set:
 [0 0 0 0 1 1 0 0 1 1 0 1 1 1 1 0 1 1 1 1 1 1 1 1 0 1 0 0 0 0 1 1 1 1 1 0 0
 1 1 0 1 1 1 1 1 0 0 1 0 1 0 1 1 0 1 0 0 1 1 0 1 1 1 0 1 0 1 1 1 1 1 1 0 1
 1 0 1 0 1 1 0 1 1 1 1 0 1 1 0 1 1 1 0 0 0 1 0 1 1 0 1 1 0 1 1 0 0 0 1 0 1
 0 0 1 1 1 1 0 0 1 1 1 1 0 1 0 0 0 1 0 1 0 1 1 1 1 1 1 1 1 0 1 1 0 0 1 0 0
 1 1]
```

   a. The numbers in this image represent each row of the testing set.
   b. For each row, there is either a 0 or a 1, which represents the values of the 'HeartDisease' label.

7. I then ran another Confusion Matrix confirmation:



   a. Though this model is slightly less accurate in terms of true negatives, its accuracy with false negatives is higher, and also produced more false positive cases and true positive cases.

Conclusion

Through using Supervised Learning via Decision Trees and Confusion Matrices, it became evident that the most reliable factor in predicting Heart Disease cases is the presence of symptoms, specifically Exercise Angina.

For this portion of the analysis, I produced two different decision trees and corresponding confusion matrices. From a clinical perspective, the second decision tree is the superior and more effective model. It resulted in a higher number of true positive cases and fewer false negative cases. While it resulted in fewer true negative cases and a higher number of false positives, it was very effective in maximizing true positives and minimizing false negatives. In a diagnostic and clinical setting, failing to identify a sick patient is a far greater risk than further investigating a healthy one. These false positive cases are significant, as they may represent "at-risk" patients who are not currently diagnosed with Heart Disease, but possess the metrics that may mimic a diseased profile. This highlights the importance of proactive monitoring and clinical care.

In the next section, I will utilize Unsupervised Learning (K-Means Clustering) to explore these "at-risk" profiles more deeply. K-Means may allow us to see if these patients can be identified through other factors, such as metabolic markers, that were lower on the importance scale in building the Decision Tree model. This would allow us to more effectively predict and identify profiles that are prone to the Heart Disease condition, and possibly even aid with diagnosis.

Unsupervised Learning with K-Means Clustering

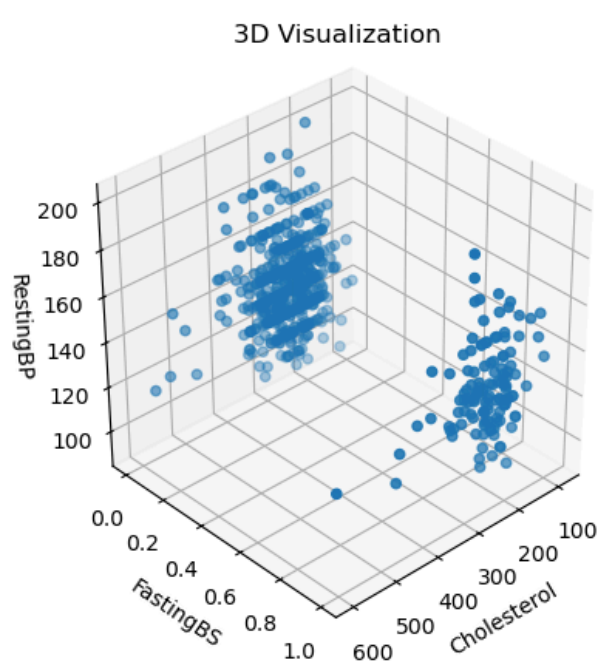
K-Means Preparation:

1. The purpose of this section is to use K-Means clustering to find if we can distinguish all existing patients in the dataset into different pattern based groups concerning their attributes and symptoms.
2. In order to prepare the cleaned dataset for K-Means calculations, all non-quantitative fields as well as the dataset label were removed.
3. At this point, the dataset looks as follows:

K-Means Table

Age	Sex	Resting BP	Cholesterol	Fasting BS	Max HR	Exercise An..	Oldpeak
40	0	140	289	0	172	0	0
49	1	160	180	0	156	0	1
37	0	130	283	0	98	0	0
48	1	138	214	0	108	1	1.5
54	0	150	195	0	122	0	0
39	0	120	339	0	170	0	0
45	1	130	237	0	170	0	0
54	0	110	208	0	142	0	0
37	0	140	207	0	130	1	1.5
48	1	120	284	0	120	0	0
37	1	130	211	0	142	0	0
58	0	136	164	0	99	1	2
39	0	120	204	0	145	0	0
49	0	140	234	0	140	1	1
42	1	115	211	0	137	0	0
54	1	120	273	0	150	0	1.5
38	0	110	196	0	166	0	0

4. I then visualized this dataset using RestingBP (resting blood pressure), FastingBS (fasting blood sugar) and Cholesterol as the axes. With this visualization, we will be able to see the configuration of all datapoints within the dataset in which we can find underlying patterns that would form cluster groups. The visualization is as follows:



K-Means Calculation:

1. To start the process of calculating K-Means for the dataset, I used the 8 columns of the dataset that were retained.

2. Then, I set the clustering value of K = 4.
   a. After experimenting with K values, I found that with K = 4, there was enough variation within the dataset to divide all data points into separate and distinct groups before the groups started to coincide with each other.
   b. Any value of K greater than 4 that was used caused the clusters to become muddied on the 3D chart.

3. With the 4 clusters prepared, I was able to find each row's cluster designation and attached this identification to the dataset:

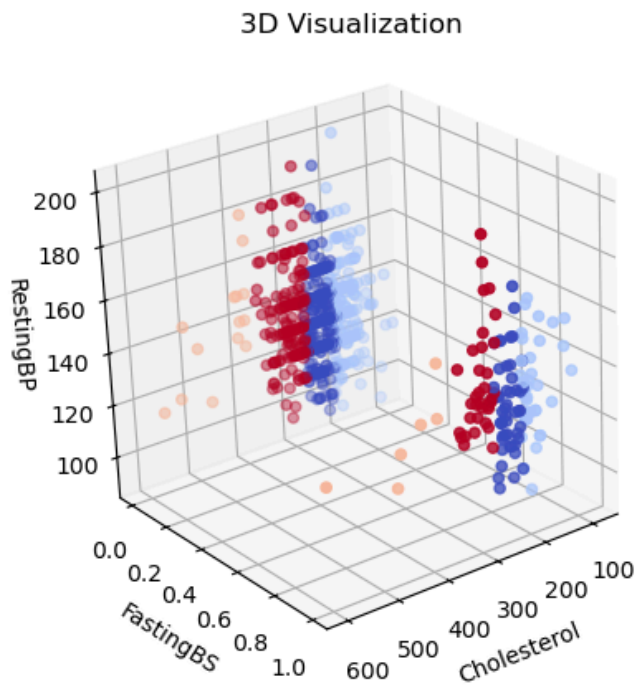
A	B	C	D	E	F	G	H	I
Age	Sex	RestingBP	Cholesterol	FastingBS	MaxHR	ExerciseAngina	Oldpeak	Cluster
40	0	140	289	0	172	0	0	3
49	1	160	180	0	156	0	1	1
37	0	130	283	0	98	0	0	3
48	1	138	214	0	108	1	1.5	1
54	0	150	195	0	122	0	0	1
39	0	120	339	0	170	0	0	3
45	1	130	237	0	170	0	0	0
54	0	110	208	0	142	0	0	1
37	0	140	207	0	130	1	1.5	1
48	1	120	284	0	120	0	0	3
37	1	130	211	0	142	0	0	1
58	0	136	164	0	99	1	2	1
39	0	120	204	0	145	0	0	1
49	0	140	234	0	140	1	1	0
42	1	115	211	0	137	0	0	1
54	1	120	273	0	150	0	1.5	3

4. I was also able to calculate the centroids for each of the 4 clusters of the dataset:

Clusters

Cluster ID	Age	Sex	Resting BP	Cholesterol	Fasting BS	Max HR	Exercise Angina	Oldpeak
0	52.2	0.2	130.4	239.1	0.2	144.6	0.3	0.9
1	52.6	0.2	131.9	186.0	0.2	137.1	0.3	0.8
2	53.9	0.5	139.6	443.8	0.3	135.9	0.4	1.0
3	54.2	0.3	137.5	299.5	0.2	137.8	0.5	1.0

5. Finally, I visualized the dataset again, but this time, with color coding to represent the separate groups based on patterns within the data



Conclusions:

From utilizing K-Means Clustering, I was able to identify 4 distinctive groups that existed within the dataset. As we can see from the cluster chart, the descriptions of each clusters are as follows:

- Cluster 0: Slightly elevated levels of cholesterol
- Cluster 1: Lowest levels of cholesterol
- Cluster 2: Highest levels of cholesterol
- Cluster 3: Elevated levels of cholesterol.

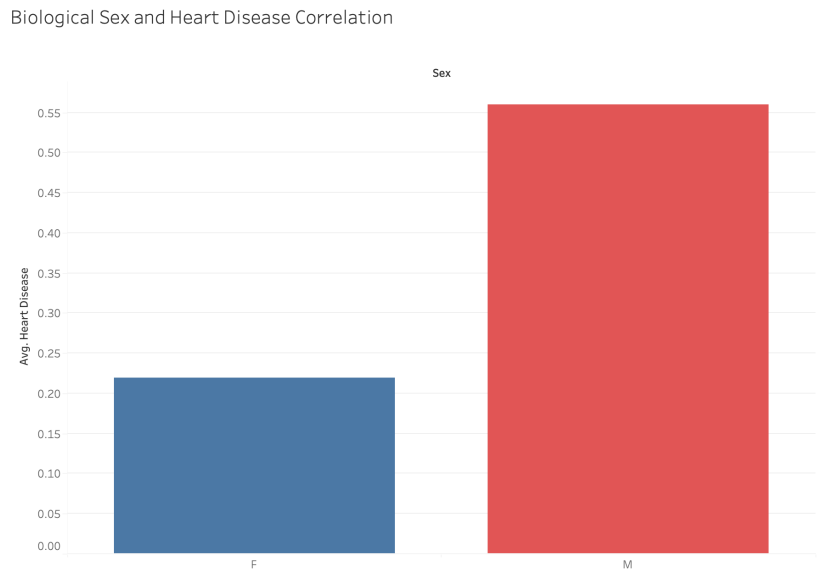
One thing to note from this clustering is that there is not much variation between the symptomatic aspects of Heart Disease, namely, Exercise Angina. While the levels of Cholesterol range from 186.0 to 443.8, there is only a difference in 0.2 for Exercise Angina in terms of the centroids for the clusters. This shows that internal metabolic markers are a far more sensitive way to group patients rather than physical pain, and also shows that metabolic conditions can nearly more than double in severity without any likelihood of an increase of symptoms. This also suggests that for the average individual, the lack of presence of symptoms such as exercise angina is not a good indicator of cardiovascular damage. An individual may not experience any chest pain, but they may be very metabolically and cardiovascularly unhealthy. These findings show that, while the Supervised Model viewed Exercise Angina as the main driver of predicting Heart Disease in patients, serum Cholesterol levels may be the key to identifying the "at-risk" false positive profiles, and even give us insight into profiles that were discovered to be false negatives as well.

In the next section of my analysis, I will be conducting Exploratory Data Analysis through SQL to gain more insight into each biological marker and what may contribute to false positive or false negative profiles.

Exploratory Data Analysis with SQL

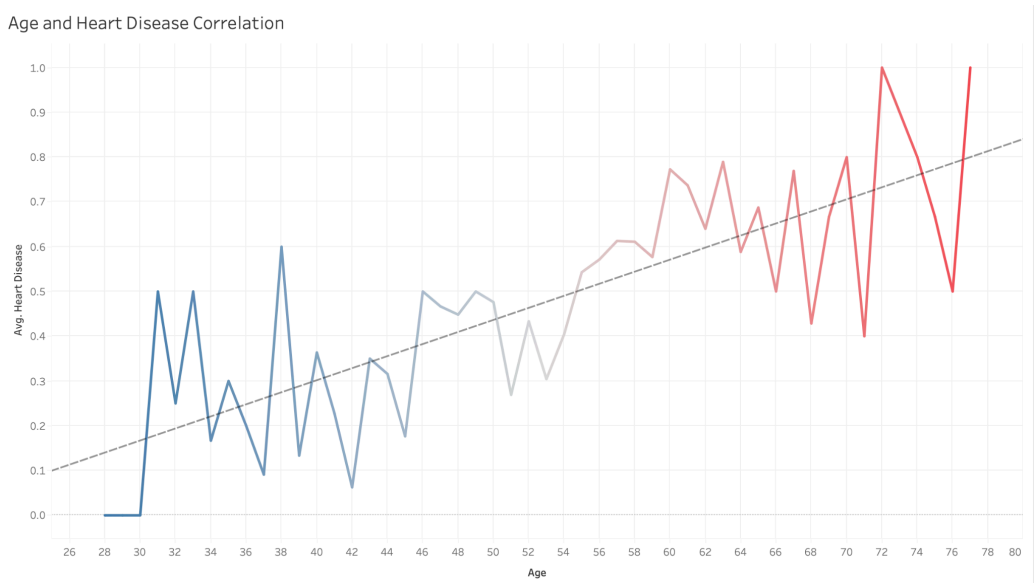
Patient Demographic Attributes:

Biological Sex and Heart Disease



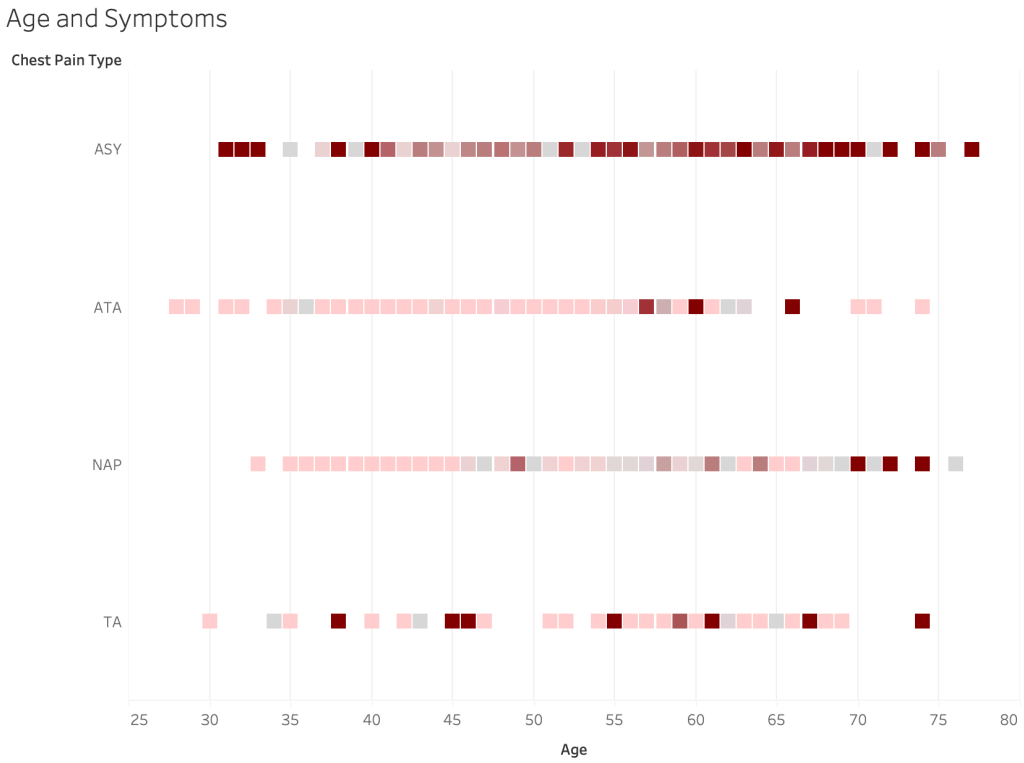
- From this chart we can see that heart disease is more prevalent in males as compared to females.
- However, biological sex on its own is not enough to determine the likelihood or risk of heart disease in a patient.

Age and Heart Disease



- This image shows that there is a greater prevalence of heart disease present in the older people in the population sample.
- According to this data, this means that there is a positive correlation between age and heart disease.
- As an individual's age increases, so does the likelihood and risk of heart disease.
- However, it is still too early to determine whether age is the primary importance among other factors when it comes to the condition.

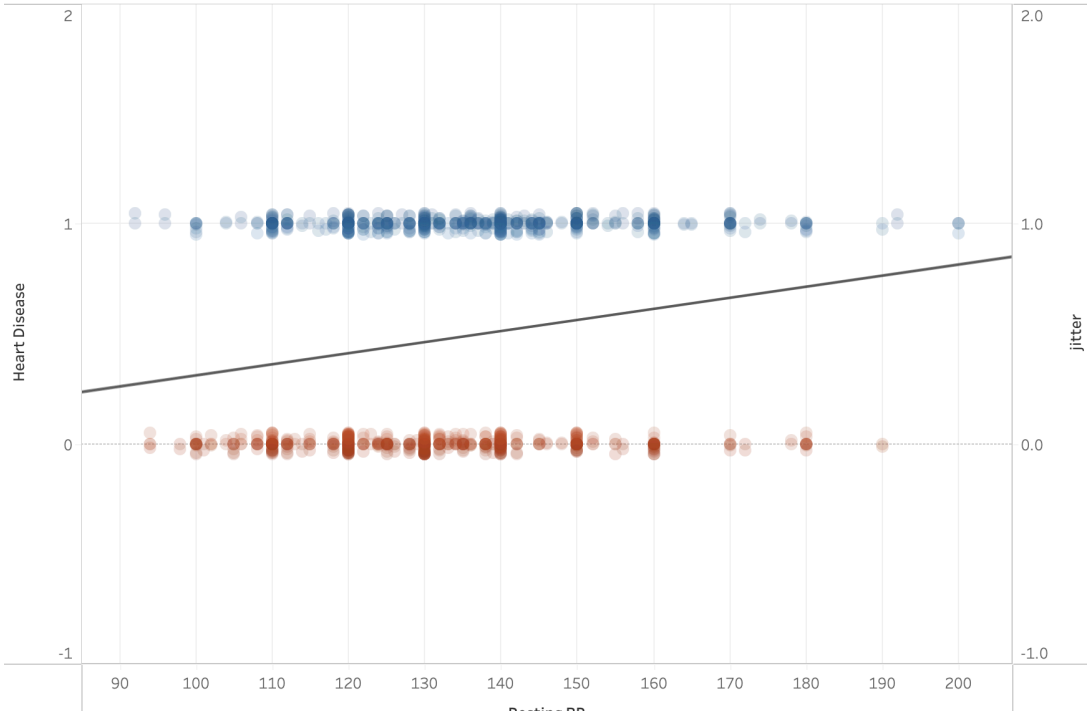
Age and Symptoms



- In this image, we see that there is a prevalence of heart disease among individuals of all ages who are asymptomatic in terms of chest pain.
- However, we can see with the other chest pain types (ATA - Atypical, NAP - Non Anginal Pain, TA - Typical Angina) older individuals show higher rates of Heart disease presence.
- So far, we can tell that a combination of Chest Pain and older age can be a sign of heart disease.

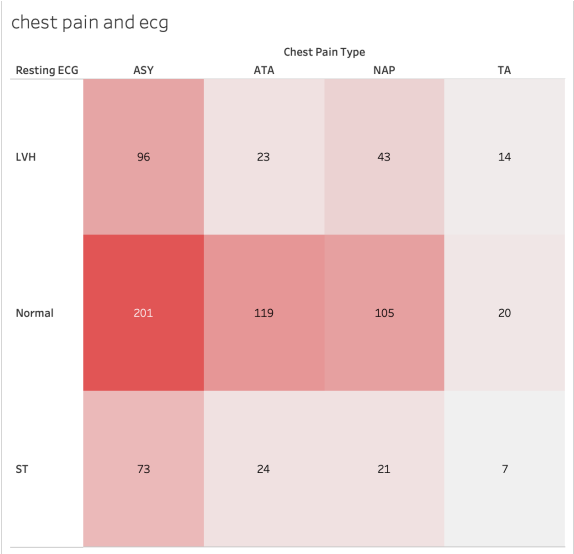
Patient Anthropometric Measurements:

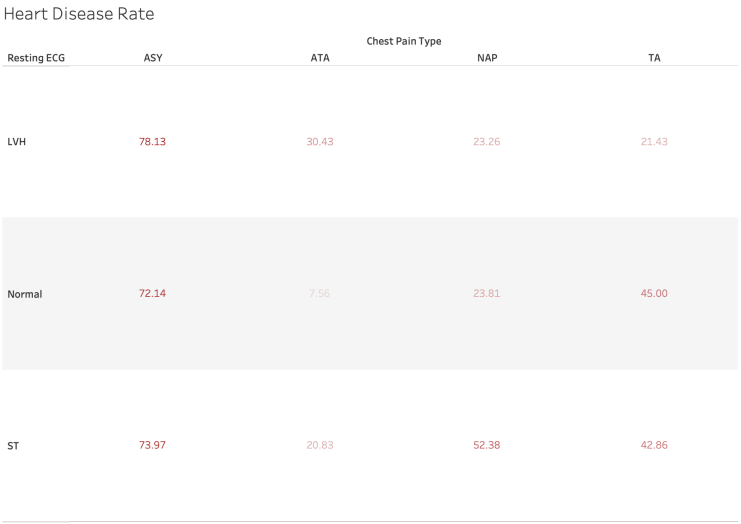
Blood pressure and Heart Disease



- In this image, we see a more defined positive relationship with blood pressure classes and heart disease.
- At each blood pressure tick mark, there is a cloud of dots, representing a number of people at that specific reading.
- This image also depicts a trend line, which shows that as blood pressure increases, the likelihood and probability of heart disease also increases.

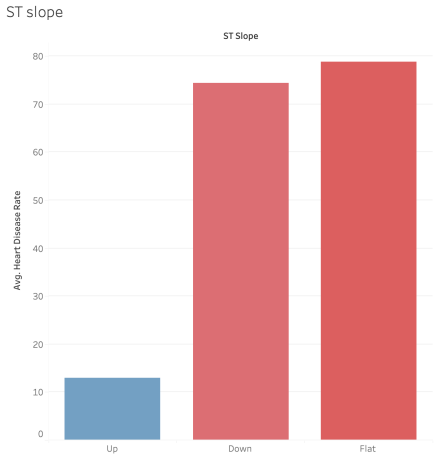
Electrocardiogram Readings and Chest Pain Types

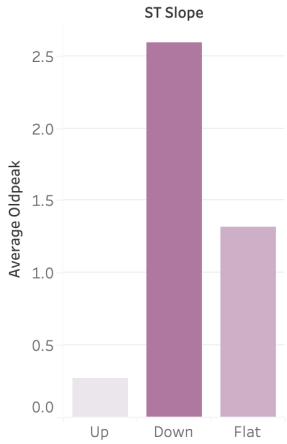


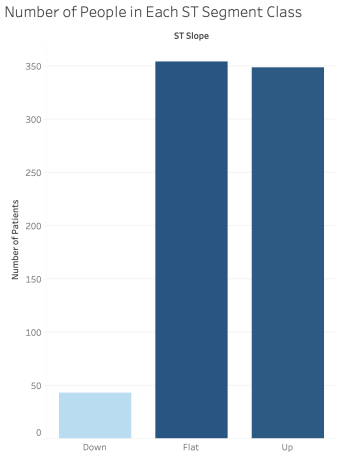


- In this heatmap, we can see the correlations between chest pain types and resting ECGs.
- A concerning aspect is that for the most part, people with any chest pain type will present with a normal ecg reading.
- This means that ecg readings on their own are not a reliable diagnostic tool for any existing heart problems.
- This graphic on the right shows that 72.14 percent of patients that presented asymptomatic with chest pain and normal ecg readings had heart disease.
- Around 24% of NAP chest pain patients with normal ecg readings had heart disease, 45% of TA chest pain patients with normal ecg readings had heart disease, though 7.56% of patients with ATA with normal ecg readings had heart disease.

ST Slope Reading and Heart Disease



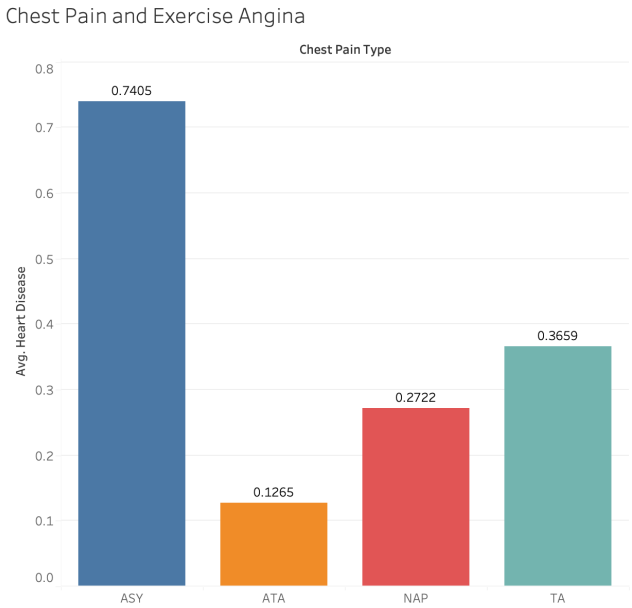


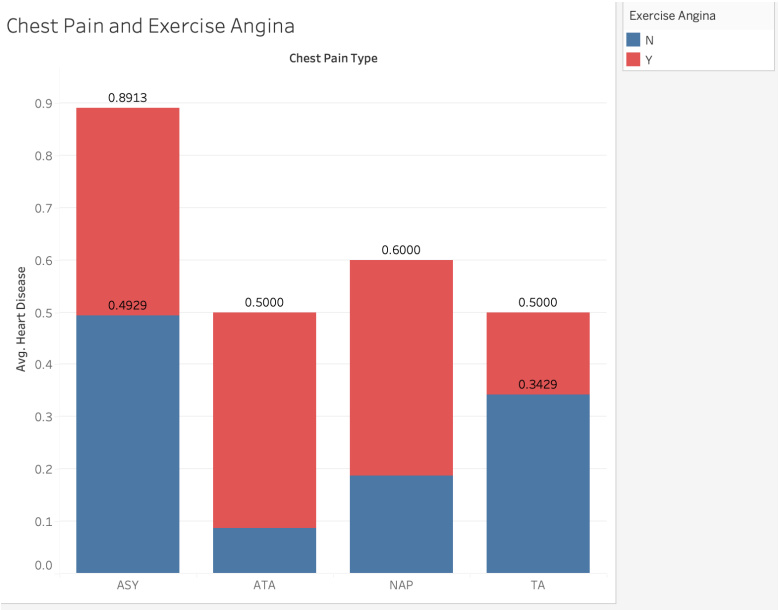


- There are three categories of ST segments: Down, Flat and Up. .
    - Up: generally represents a healthy heart recovery
    - Flat: The heart is struggling or taking too long to reset.
    - Down: The heart is failing to reset properly due to lack of oxygen.
- We can see from this image is that patients that present with abnormal ST segments also present with high rates of heart disease, allowing us to infer that abnormal ST segments are a good indicator of heart disease risk and presence.
- We can also see that each ST slope category correlates with an average old peak value, where the higher the old peak value is, the more dangerous the situation generally is.
- Something to note: higher rates of heart disease in patients presenting with flat ST segments as compared to down segments, even though the ecg reading is not as severe. This could be explained by the amount of patients in the data collection that fall within those categories.

Patient Symptoms and Heart Disease:

Chest Pain and Exercise Angina



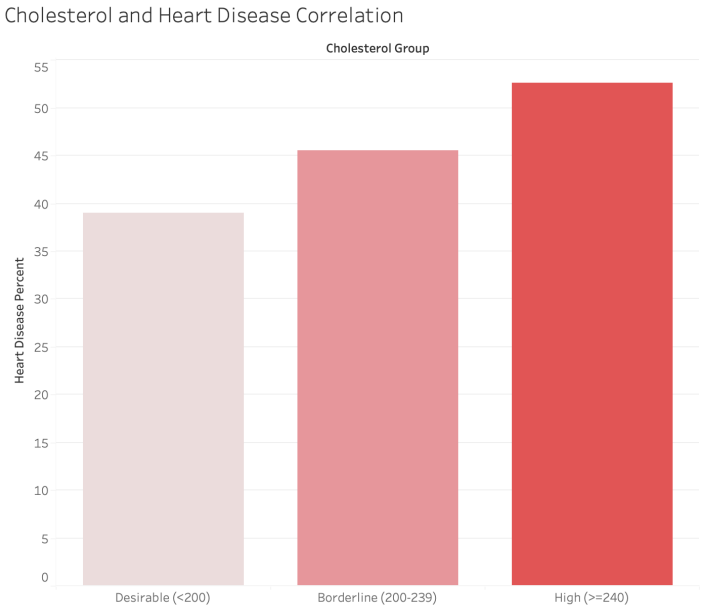


- In the chart on the left, we see the prevalence of heart disease in patients with certain chest pain types.
  - For asymptomatic patients, around 74% of patients were recorded to have heart disease. For Atypical chest pain patients, around 12.7% of patients were recorded to have heart disease. F
  - or Non Anginal chest pain patients, around 27.2% of patients were recorded to have heart disease.
  - For Typical Angina chest pain patients, around 36.6% of patients were recorded to have heart disease.

- However, when we look to the chart on the right, we see that when chest pain is present in conjunction with exercise angina (chest pain with exercise), the rates of heart disease increases significantly.
  - ATA - jumps to 50% NAP - jumps to 60% TA - jumps to 50%
  - This shows that chest pain with exercise angina is a significant element of prediction for heart disease in individuals.
    - This matches what we discovered as the main driver for Heart Disease prediction during the Supervised Learning portion of this analysis.

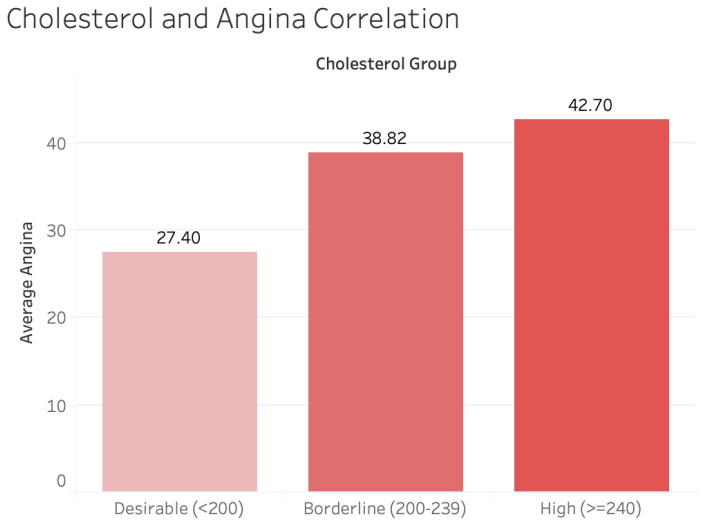
Patient Metabolic Markers:

Cholesterol and Heart Disease



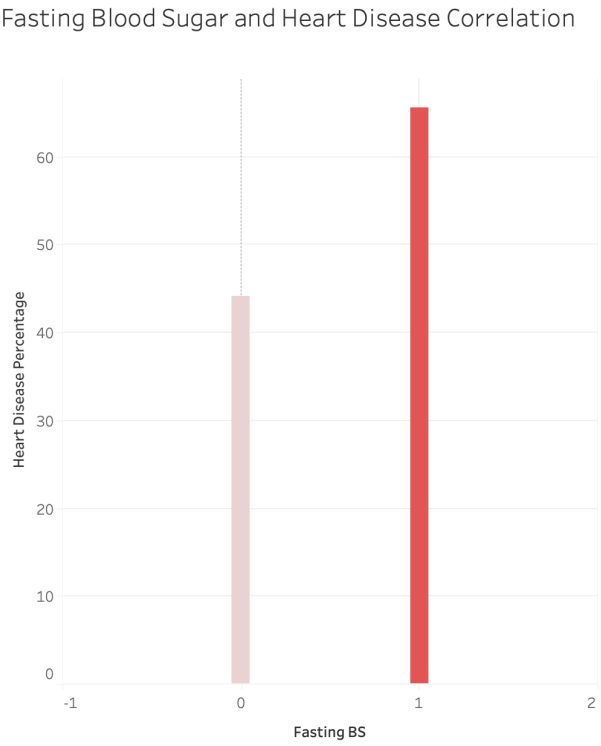
- In this image, there are 3 categories for cholesterol levels.
  - <200 mg/dL signify normal/desirable levels.
  - 200-239 mg/dL signify borderline high levels of cholesterol.
  - Levels of 200mg/dL or more are cases of high cholesterol content.
- Each color in this graph represents the presence of heart disease in the patients at these cholesterol level groups.
- We can see that as cholesterol content increases in patients, so does the likelihood of heart disease.

Cholesterol and Exercise Angina



- Cholesterol has a clear positive correlation with Exercise Angina, where the higher the cholesterol, the more likely one will experience heart pain when exercising.
- However, even the highest cholesterol group, despite the upward trend, only shows exercise Angina less than half of the time.
- Exercise Angina was our primary importance factor in our decision tree, which starts to reveal that there may be other factors at play, especially for those who present normally but have the heart disease condition, and also for the false negatives confirmed in our confusion matrix.

Fasting Blood Sugar and Heart Disease



- In this chart, there are two groups: 0 representing lower fasting blood sugar levels(< 120 mg/dl) and 1 representing higher levels of fasting blood sugar (> 120 mg/dl)
- There is a similar trend of heart disease prevalence with patients presenting with normal levels of fasting blood sugar as opposed to high levels of fasting blood sugar.
- Patients with higher levels of fasting blood sugar tend to have higher rates of heart disease.

Midway Conclusions:

- Each factor on its own (age, sex, ecg readings, chest pain and angina, cholesterol, and fasting blood sugar) all seem to increase the risk of heart disease in an individual as the conditions become more severe.
- Because of this, it may be more useful as a diagnostic tool and preventative measure to analyze those who present normally in terms of ecg readings and presence of any chest pain or exercise angina.
- This would help us reveal what may be happening with the "at-risk" false positive profiles and possibly even the false negative profiles in our confusion matrix.

Multiple Variables Analysis

Normal patient (Normal Readings, Normal Metabolic Markers, No Exercise Angina)

Resting ECG	Chest Pain Type: ASY	ATA	NAP	TA
LVH	0.7500	0.0000	0.7500	0.0000
Normal	0.3000	0.0000	0.1364	0.3333
ST	0.3333	0.0000	0.0000	0.0000

- In this image, we can see the rates of heart disease in patients of various chest pain types and ecg readings combined.
- The results of this table have also been filtered to exclude the presence of exercise angina.
- We now turn our attention to the middle row of the table, specifically ASY (asymptomatic) x Normal.
    - Patients who are asymptomatic and have normal ECG readings as well as normal cholesterol and fasting blood sugar levels present with a rate of 30% when it comes to heart disease.

Normal patient but with High Cholesterol

Resting ECG	Chest Pain Type: ASY	ATA	NAP	TA
LVH	0.640	0.300	0.063	0.500
Normal	0.462	0.080	0.185	0.333
ST	0.167	0.000	0.000	1.000

- In this image, the results have been filtered to only people in the high cholesterol group.
- We can see in the table, the patients in the normal category, being normal ECG readings and Asymptomatic in terms of any chest pain, the rate of Heart Disease has jumped to 46.2%

Normal but with High Cholesterol and High Fasting Blood Sugar

Resting ECG	Chest Pain Type: ASY	ATA	NAP	TA
LVH	0.500	1.000	0.000	0.500
Normal	0.750	0.000	0.250	0.667
ST	1.000	1.000		

- Now the results have been tailored to people with both high cholesterol and higher fasting blood sugar levels, all while presenting normally in terms of ECG readings and any chest pain.
- The rate of heart disease in these patients has jumped to 75%!

Final Conclusions:

In the Supervised Learning phase of this analysis, the primary driver for predicting the presence of Heart Disease was found to be symptomatic presentation, specifically Exercise Angina. While the final model demonstrated high overall accuracy, the results yielded a notable number of False Positives and a smaller subset of False Negatives. I determined that these misclassifications were not merely statistical errors, but rather a reflection of the complexity of cardiovascular health. The False Positive cases, in particular, may represent patients who, despite not having a confirmed Heart Disease diagnosis, possess physiological risk factors that the model recognizes as diseased. These patients may be occupying a clinical gray area where their metabolic metrics (such as Cholesterol or Blood Pressure) mimic those of positive cases, highlighting the transition from "healthy" to "at-risk."

To investigate these misclassifications, I applied K-Means Clustering to identify any existing patterns within the dataset that could explain the underlying risk profiles of these cases. Surprisingly, the cluster groups that were formed were not based on variance with the presence of Exercise Angina in patients, but rather were formed on the basis of their serum cholesterol levels. This demonstrated that, even if a patient may not be experiencing any Exercise Angina, which was the driving factor for Heart Disease prediction in our Supervised Learning Decision Tree model, their metabolic panel could present the same way a diseased and symptomatic individual would, highlighting the discrepancy between true positive and false positive cases.

To see whether this was the case, I used Exploratory Data Analysis using SQL to investigate this issue further. I first explored the impact of each individual biological marker. Consistent with the results of with the Decision Tree model, the EDA results confirmed that Exercise Angina maintained the strongest positive correlation with a Heart Disease diagnosis.However, unlike what we saw with our Decision Tree model, EDA showed that as each variable reached more extreme values, the frequency of Heart Disease also increased. Though these single variable queries confirmed the general conclusions of the machine learning analysis, they did not provide any additional insights into what could cause false positive or negative cases.

However, upon exploring patients in the dataset who presented normally and filtered specifically to exclude Exercise Angina, I found that the rate of Heart Disease increased significantly when Cholesterol levels were adjusted to “High” and escalated even further when Fasting Blood Sugar was adjusted to “Elevated.” This shows that even if a person presents as symptomatically normal (with no Chest Pain or Exercise Angina), the likelihood of a patient being diagnosed with Heart Disease increases significantly if both their Cholesterol and Fasting Blood Sugar levels are high. This can explain both the false positive and negative profiles in our Decision Tree model, and explain why our clustering was grouped based on Cholesterol levels.

To conclude, the main driver of predicting Heart Disease in a patient is if they present with Exercise Angina. However, high cholesterol and fasting blood sugar levels are also causes for concern, even if the patient is outwardly asymptomatic.